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COURSE NAME : ANALOG AND DIGITAL COMMUNICATION (ECA0802)

EX NO: 9	SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF ENCODED OUTPUT BIT SEQUENCES USING MATLAB
DATE:	

Objectives

- (i) To implement convolutional encoding for a given binary input sequence using a rate $\frac{1}{2}$ convolutional encoder and tabulate the corresponding encoded output pairs.
- (ii) To visualize and compare the input bit sequence and encoded output bit sequence using MATLAB plots to analyze the redundancy introduced by convolutional coding.

MATLAB code

```
clc;
clear;
close all;
% Fixed input sequence
input_bits = [1 0 0 1 1];
% Convolutional encoder
trellis = poly2trellis(3,[7 5]);
encoded_bits = convenc(input_bits,trellis);
% Separate encoded bits into pairs
encoded_pairs = reshape(encoded_bits,2,[]);
% Create output strings
for i = 1:length(input_bits)
    output_pair{i} = sprintf('%d%d', ...
        encoded_pairs(i,1),encoded_pairs(i,2));
end
% Create table
```

```

Bit_Position = (1:length(input_bits));
Input_Bit = input_bits';
Output_Encoded = output_pair';
T = table(Bit_Position,Input_Bit,Output_Encoded);
disp('Convolutional Encoding Table');
disp(T);
% Plot
figure;
subplot(2,1,1)
stairs(input_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Input Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on
subplot(2,1,2)
stairs(encoded_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Encoded Output Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')

```

Procedure

1. Open MATLAB and create a new script file.
 2. Initialize the MATLAB workspace using `clc`, `clear`, and `close all`.
 3. Define the input bit sequence as `[1 0 0 1 1]`.
 4. Create a convolutional encoder trellis using a constraint length of 3 and generator polynomials `[7 5]`.
 5. Encode the input sequence using the `convenc()` function.
 6. Group the encoded bits into pairs corresponding to each input bit.
 7. Create and display a table containing the bit position, input bit, and encoded output pair.
 8. Observe the encoded output pairs generated by the encoder.
- `grid on`

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To implement convolutional encoding for a given binary input sequence using a rate $\frac{1}{2}$ convolutional encoder and tabulate the corresponding encoded output pairs.

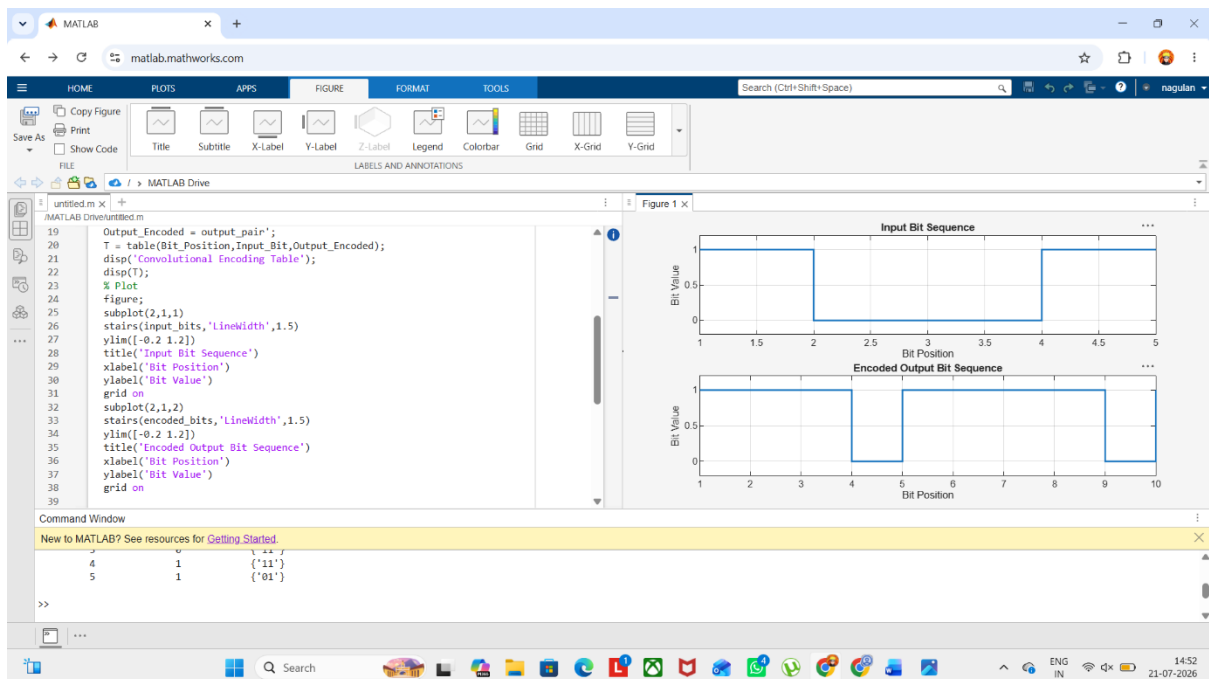
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Tabulation

Bit position	Input bit	Encoded sequence
1	1	11
2	0	10
3	0	11
4	1	11
5	1	01

OUTPUT



Result

The convolutional encoder was successfully implemented using a constraint length of 3 and generator polynomials $[7 \ 5]$. The input bit sequence $[1 \ 0 \ 0 \ 1 \ 1]$ was encoded into the output sequence 1110111101 , and the corresponding encoded output pairs were tabulated successfully.